IGCSE Physics 0625

2025 Compact Workbook · 10 multiple choice + 3 theory questions per topic

Name: _____

Class: _____

Cambridge IGCSE Physics 0625 — 2025 (Feb/Mar, May/Jun, Oct/Nov)

练习册

Contents

Topic · Subtopic · page

1 MOTION, FORCES AND ENERGY

1.1 Physical quantities and measurement techniques
1.2 Motion
1.3 Mass and weight
1.4 Density
1.5 Forces
1.6 Momentum
1.7 Energy, work and power
1.8 Pressure

2 THERMAL PHYSICS

2.1 Kinetic particle model of matter
2.2 Thermal properties and temperature
2.3 Transfer of thermal energy

3 WAVES

3.1 General properties of waves
3.2 Light
3.3 Electromagnetic spectrum
3.4 Sound

4 ELECTRICITY AND MAGNETISM

4.1 Simple phenomena of magnetism
4.2 Electrical quantities
4.3 Electric circuits
4.4 Electrical safety
4.5 Electromagnetic effects

5 NUCLEAR PHYSICS

5.1 The nuclear model of the atom
5.2 Radioactivity

6 SPACE PHYSICS

6.1 The Earth and the Solar System
6.2 Stars and the Universe

1 MOTION, FORCES AND ENERGY

13 questions · Paper 2 & Paper 4

1 A student determines the diameter of a metal ball.

Which apparatus is needed to get an accurate measurement?

A a ruler and a protractor

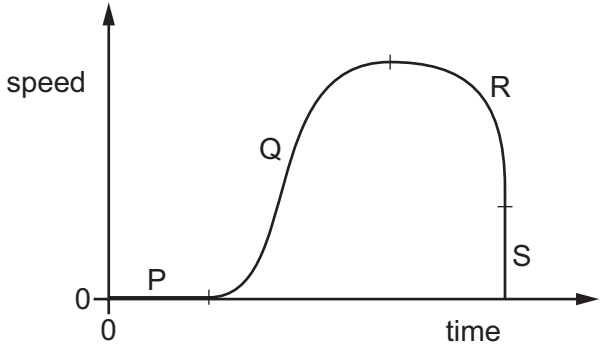
B a ruler only

C a protractor and a set square

D a ruler and two rectangular blocks

Date: _____ Score: _____

1 The speed–time graph for a racing car is divided into four sections, P, Q, R and S. The car starts the race but soon crashes into a wall of tyres.



Which sections of the graph show that there is an acceleration that is changing?

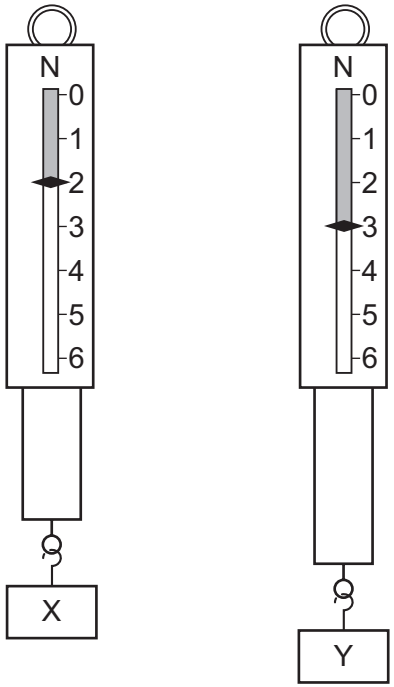
A P, Q, R and S

B Q, R and S only

C Q and R only

D R and S only

3 The diagrams show two metal blocks, X and Y, suspended from identical force meters. X and Y have identical dimensions.



Which statement about X and Y is correct?

A They have different volumes and different weights.

B They have different volumes and equal weights.

C They have equal volumes and equal weights.

D They have equal volumes and different weights.

4 A girl has a gold necklace.

Which apparatus does she need to find the density of the gold?

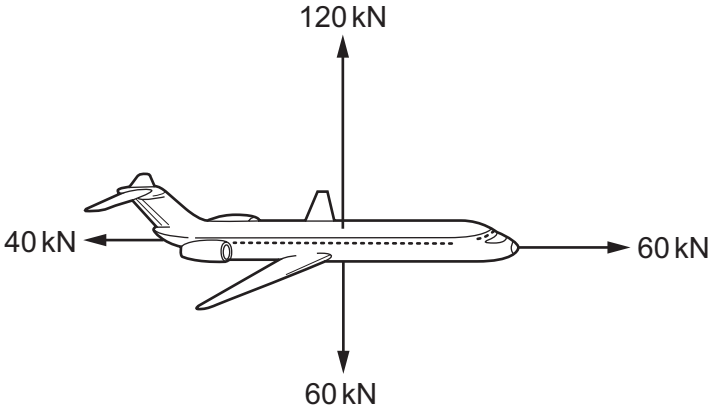
	balance	measuring cylinder	thermometer	ruler
A	✓	✓	✓	✓
B	✓	✓	✓	✗
C	✓	✓	✗	✗
D	✓	✗	✗	✗

key

✓ = needed

✗ = **not** needed

2 The diagram shows the four forces acting on an aircraft in flight.



Which arrow shows the direction of the resultant of the four forces?

A



B



C



D



7 Which statement is **not** correct?

A the momentum of a system is conserved in all interactions

B impulse = change of momentum

C force acting = change of momentum × time

D momentum = mass × velocity

8 A machine is very efficient.

What does this mean?

A It produces a large amount of power.

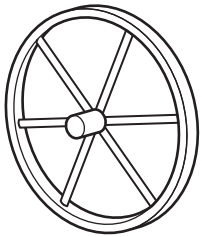
B It uses very little energy.

C It wastes very little energy.

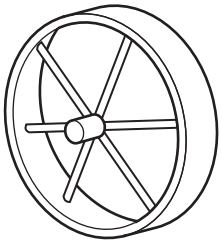
D It works very quickly.

9 A farmer has two carts. The carts have the same weight, but one has four narrow wheels and the other has four wide wheels.

narrow wheel



wide wheel



Which cart wheels sink less into soft ground and what is the reason?

	cart wheels	reason
A	narrow	greater pressure on the ground
B	narrow	less pressure on the ground
C	wide	greater pressure on the ground
D	wide	less pressure on the ground

Date: _____ Score: _____

1 Which instrument is most suitable to determine the volume of a small irregularly shaped stone?

A 30 cm ruler

B digital timer

C measuring cylinder

D tape measure

2 An athlete runs at a speed of 8 m / s for 10 s, and then at a speed of 6 m / s for 12 s.

Which calculation gives the average speed of the athlete in m / s?

A $\frac{8+6}{2}$

B $\frac{(8\times10)+(6\times12)}{22}$

C $\frac{(8\div10)+(6\div12)}{22}$

D $\frac{(10\div8)+(12\div6)}{22}$

1 **(a)** Circle the vector quantities in the list.

acceleration **mass** **speed** **time** **velocity**

[1]

(b) Fig. 1.1 shows the speed–time graph for a train travelling from station A to station B.

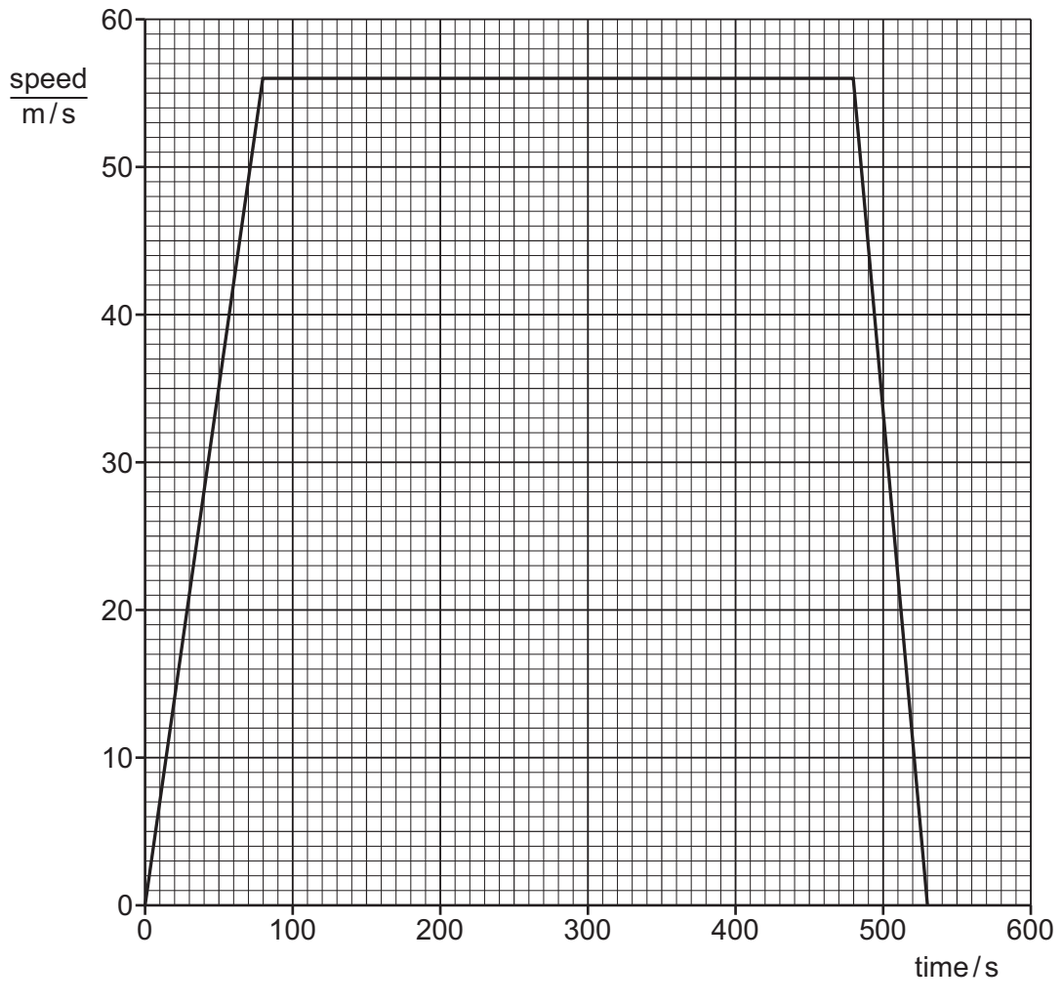


Fig. 1.1

(i) State the maximum speed of the train.

maximum speed = ... [1]

(ii) Describe the motion of the train between station A and station B.

.....

.....

..... [2]

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

© UCLES 2025

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(iii) Calculate the distance between station A and station B.

distance = ... [3]

(iv) On a different day, the train takes 650 s to travel between station A and station B.

Suggest **one** change to the motion of the train that leads to this longer journey time.

.....

..... [1]

[Total: 8]

2 Table 2.1 contains information about the planet Mars.

Table 2.1

mass	6.4×10^{23} kg
gravitational field strength at surface	3.7 N / kg
average density	3900 kg / m^3

(a) Define gravitational field strength.

.....

..... [1]

(b) **(i)** An object has a weight of 42 N at the surface of the Earth.

Calculate the weight of the object at the surface of Mars.

weight = ... [2]

(ii) Calculate the volume of Mars.

volume = ... [2]

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

© UCLES 2025

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(c) Fig. 2.1 shows a space buggy that is tested on Earth. The buggy is travelling at a constant speed in a straight line. The driving force on the buggy is 30 N.

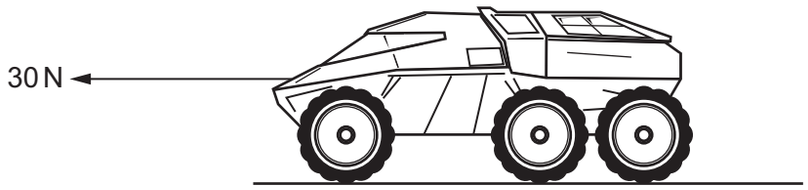


Fig. 2.1

(i) Draw and label **one** arrow on Fig. 2.1 to show the size and direction of the resistive forces on the buggy. [2]

(ii) Air resistance on Mars is less than air resistance on Earth.
The same driving force, 30 N, is exerted on the buggy on Mars.

1. State the effect this has on the resultant force on the buggy on Mars.

.....

2. State the relationship between resistive forces, driving force and resultant force.

.....
[1]

[Total: 8]

2 **(a)** A resultant force is applied to an object moving with a velocity v in a straight line.

(i) State **two** different changes to the motion that the resultant force may cause.

1 .....

2 .....
[2]

(ii) State **one** other way that forces may change a stationary object.

..... [1]

(b) Describe how a uniform metre ruler, a pivot and a selection of masses can be used to demonstrate that there is no resultant moment on an object in equilibrium.

You may include a labelled diagram in your answer.

.....

.....

.....

.....

.....

..... [4]

[Total: 7]

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

2 THERMAL PHYSICS

13 questions · Paper 2 & Paper 4

10 A hot gas and a cool liquid are at the same pressure.

Which row describes the separation and the motion of the particles of a hot gas compared to those of a cool liquid?

	separation of gas particles	motion of gas particles
A	greater than a cool liquid	faster than a cool liquid
B	greater than a cool liquid	slower than a cool liquid
C	less than a cool liquid	faster than a cool liquid
D	less than a cool liquid	slower than a cool liquid

12 The specific heat capacity of solid P is greater than that of solid Q.

What does this statement mean?

A Less energy is needed to raise the temperature by 1 °C of unit mass of solid P than unit mass of solid Q.

B Less energy is needed to melt unit mass of solid P than unit mass of solid Q.

C More energy is needed to raise the temperature by 1 °C of unit mass of solid P than unit mass of solid Q.

D More energy is needed to melt unit mass of solid P than unit mass of solid Q.

15 A student has a copper rod and a wooden rod. Both rods have the same dimensions.

The student puts one end of each rod into a beaker of crushed ice.

After a few minutes, the student picks up the rods by the ends **not** in the ice.

Which statement is correct?

A The copper rod feels colder than the wooden rod because copper is a better conductor than wood.

B The copper rod feels warmer than the wooden rod because copper is a better conductor than wood.

C The wooden rod feels colder than the copper rod because wood is a better conductor than copper.

D The wooden rod feels warmer than the copper rod because wood is a better conductor than copper.

11 A fixed mass of gas has a volume of 1.2 m^3 and is at a pressure of 200 kPa. The gas expands at constant temperature to a volume of 1.8 m^3.

What is the new pressure of the gas?

A 120 kPa **B** 130 kPa **C** 300 kPa **D** 600 kPa

13 When water evaporates, what escapes from the surface of the water?

A individual atoms

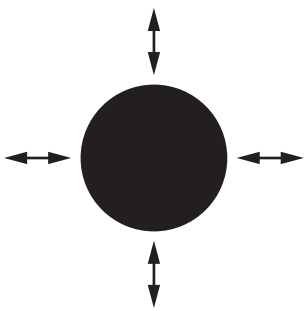
B individual molecules

C individual protons

D tiny drops of water

16 The diagram shows a black object X at constant temperature.

The arrows represent the absorption and emission of thermal radiation at the surface.



Object Y is at the same constant temperature as X and is identical except that it has a white surface.

Which row correctly compares object X with object Y?

	rate of absorption for X	rate of emission for X
A	higher	higher
B	lower	higher
C	higher	lower
D	lower	lower

14 A sealed container of gas is heated. The pressure of the gas increases.

Which statement explains this increase in pressure?

A The forces of the gas particles striking the walls of the container increase.

B The forces of attraction between the gas particles and the walls of the container increase.

C The gas particles collide with each other more frequently.

D The gas particles lose more energy when they strike the walls of the container.

12 The distance between two electricity pylons is 60 m. The cable that connects the two pylons is 62 m in length.

Why is the cable longer than the distance between the two pylons?

A to allow for contraction of the cable in cold weather

B to create a slope in the cable for electrons to flow down

C to keep the current low and the voltage high

D to reduce magnetic fields around the cable

14 Which statement about convection is **not** correct?

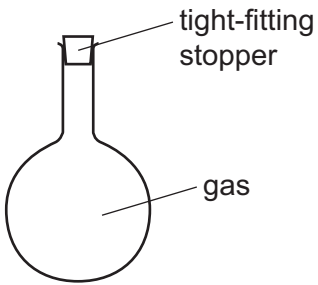
A It enables water in a pan on a cooker to get evenly heated.

B It happens in liquids and gases.

C It occurs because cooler liquids rise above warmer liquids.

D It occurs because the density of a liquid decreases when it is heated.

10 A flask with a tight-fitting stopper contains a gas. The gas and flask are initially at room temperature.



The gas and flask are cooled.

Which row describes the average speed of the gas particles and the pressure of the gas as the temperature decreases?

	average speed of the gas particles	pressure of the gas
A	decreases	decreases
B	decreases	stays the same
C	increases	increases
D	increases	stays the same

4 Fig. 4.1 shows gas trapped in a cylinder by a piston.

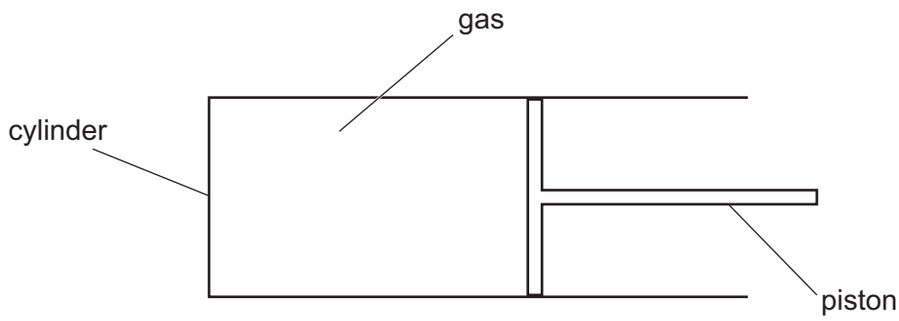


Fig. 4.1

(a) The volume of gas is 240 cm^3.
The piston is pushed to the left and is held in its new position.

(i) The pressure of the gas increases from 1.0×10^5 Pa to 1.4×10^5 Pa.

The temperature of the gas remains constant.

Calculate the volume of the gas when the piston is in its new position.

volume = ... cm^3 [3]

(ii) The area of the piston in contact with the gas is 1.9×10^{-3} m^2.

Calculate the force exerted on the piston by the gas when the piston is held in its new position.

force = ... [2]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

© UCLES 2025

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(iii) The distance moved by the piston is 0.036 m. The average force exerted by the piston as it moves is 220 N.

Calculate the mechanical work done by the piston. State the equation you use.

work done = ... [2]

(b) Explain, in terms of particles, why gases can be compressed but liquids cannot.

.....

.....

..... [1]

[Total: 8]

4 **(a)** A 12 V, 50 W immersion heater is used to heat 0.15 kg of water in a beaker. The water is initially at a room temperature of 20 °C. The specific heat capacity of water is 4200 J / (kg °C).

Calculate the energy supplied to raise the temperature of the water from 20 °C to 58 °C.

energy = ... [3]

(b) The immersion heater is removed from the beaker.

One metal rod and one plastic rod are placed in the beaker of hot water as shown in Fig. 4.1. The rods are at room temperature (20 °C) before they are placed into the beaker.

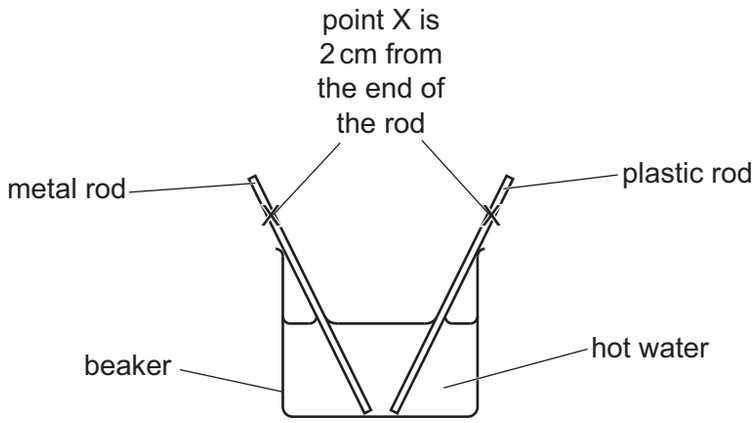


Fig. 4.1

Describe how the temperature of point X on each rod changes after the rods are placed in the beaker. Explain your answer.

.....

.....

.....

.....

..... [4]

[Total: 7]

DO NOT WRITE IN THIS MARGIN

4 Fig. 4.1 shows a heater used to warm the air in a room.

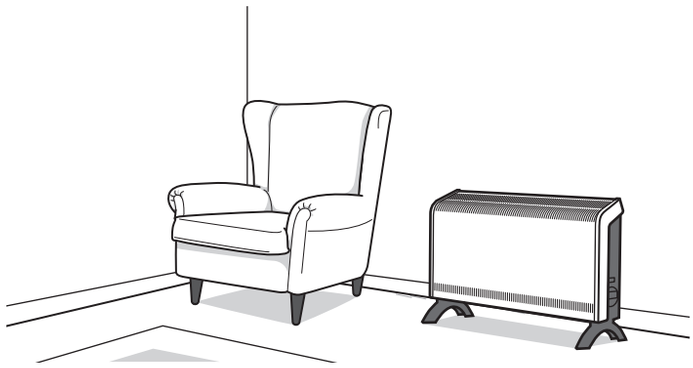


Fig. 4.1

(a) (i) State the main method of thermal energy transfer throughout the air in the room.

..... [1]

(ii) Explain how the heater warms **all** the air in the room.

.....

.....

.....

.....

..... [3]

(b) The power of the heater is 2.0 kW when it is connected to the mains supply with an e.m.f. of 230 V.

(i) Show that the current in the heater is approximately 8.7 A.

[2]

(ii) The plug connecting the heater to the mains supply is fitted with a fuse.

Fuse ratings of 3A, 5A, 10A and 13A are available.

State which fuse is used. Explain your answer.

fuse .....

explanation .....

.....

..... [2]

[Total: 8]

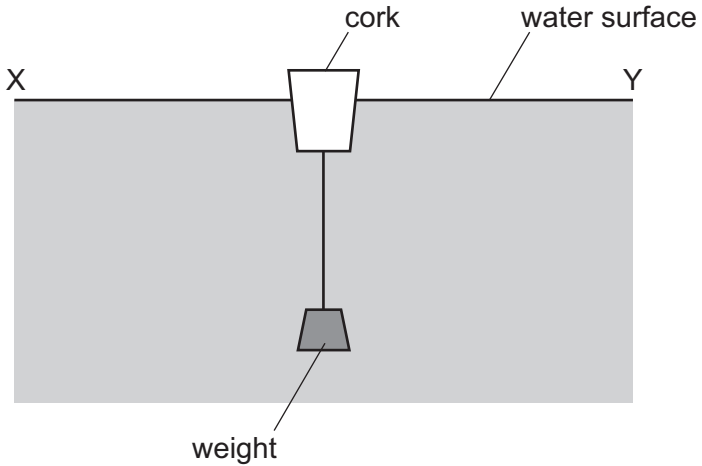
© UCLES 2025

DO NOT WRITE IN THIS MARGIN DO NOT WRITE IN THIS MARGIN DO NOT WRITE IN THIS MARGIN DO NOT WRITE IN THIS MARGIN DO NOT WRITE IN THIS MARGIN

3 WAVES

13 questions · Paper 2 & Paper 4

17 The diagram shows a cork with a weight attached so that the cork floats upright in water.



Transverse waves travel across the water from X to Y.

In which direction do the waves make the cork move?

A $\rightarrow \leftarrow$ right and left

B $\uparrow\downarrow$ up and down

C $\rightarrow$ only to the right

D $\leftarrow$ only to the left

18 A thin converging lens is used as a magnifying glass.

How does the image compare with the object?

A real and inverted

B real and upright

C virtual and inverted

D virtual and upright

22 Satellites are sometimes used to receive and re-transmit television signals.

Which row shows the type of satellite and the electromagnetic waves used?

	type of satellite	electromagnetic waves used
A	low orbit	infrared
B	low orbit	microwaves
C	geostationary	infrared
D	geostationary	microwaves

23 A sound wave travels through the air and strikes a solid block of steel.

Some of the sound is reflected from the steel, and some is transmitted through the steel.

How does the speed and the wavelength of the sound in steel compare to the speed and the wavelength of the sound in air?

	speed in steel	wavelength in steel
A	less than in air	less than in air
B	less than in air	greater than in air
C	greater than in air	less than in air
D	greater than in air	greater than in air

18 Parallel waves in a ripple tank are diffracted as they pass through a gap in a barrier.

Three changes that can be made to this arrangement are listed.

1 Decrease the wavelength of the waves.

2 Increase the amplitude of the waves.

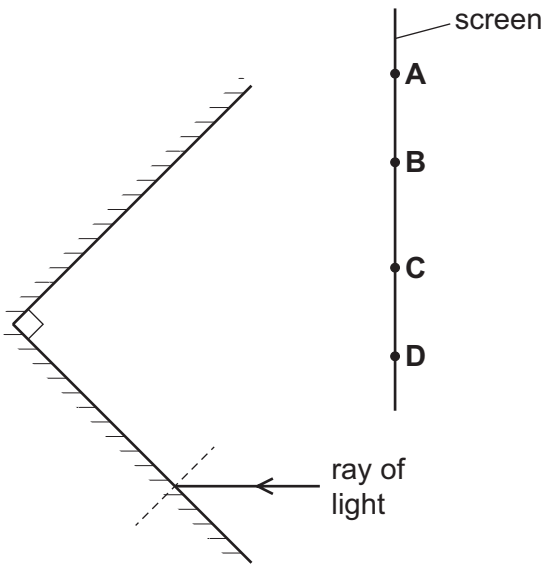
3 Decrease the size of the gap.

Which changes will cause the shape of the diffracted waves to be less curved?

A 1, 2 and 3 **B** 1 and 3 only **C** 1 only **D** 2 and 3 only

19 The diagram shows two plane mirrors at 90° to each other. A ray of light is incident on one of the mirrors. The ray reflects off both mirrors before reaching a screen.

At which labelled point does the ray reach the screen?



22 Ultraviolet radiation and infrared radiation are both parts of the electromagnetic spectrum.

However, they differ in wavelength and frequency.

Which row describes how they differ?

	ultraviolet	infrared
A	longer wavelength than infrared	higher frequency than ultraviolet
B	longer wavelength than infrared	lower frequency than ultraviolet
C	shorter wavelength than infrared	higher frequency than ultraviolet
D	shorter wavelength than infrared	lower frequency than ultraviolet

23 A police car with its siren sounding is stationary in heavy traffic. A pedestrian notices that, although the loudness of the sound produced does **not** change, the pitch varies.

Which row describes the amplitude and the frequency of the sound?

	amplitude	frequency
A	constant	constant
B	constant	varying
C	varying	constant
D	varying	varying

19 The frequency of a wave is doubled. The speed of the wave does **not** change.

What happens to the wavelength of the wave?

A It becomes four times as large.

B It does **not** change.

C It doubles.

D It halves.

20 The speed of light in a material is 50% of the speed of light in air.

What is the refractive index of the material?

A 0.5 **B** 2.0 **C** 6.0×10^8 **D** 1.5×10^8

7 **(a)** Fig. 7.1 is a scale drawing of light waves approaching a narrow slit.

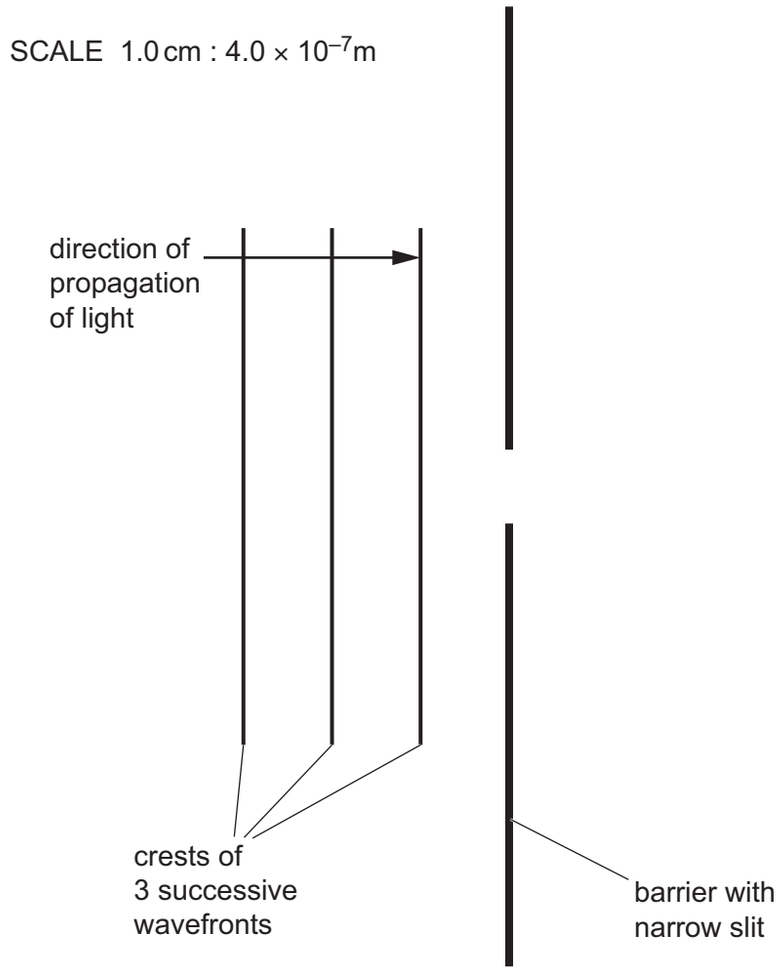


Fig. 7.1

(i) Name the wave effect produced by the narrow slit.

..... [1]

(ii) Using Fig. 7.1, determine the wavelength of the light. Give your answer to **two** significant figures.

wavelength = ... [2]

(iii) On Fig. 7.1, draw **three** wavefronts that have passed through the narrow slit. [3]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(b) A foghorn emits a sound with frequency 380 Hz. The sound is heard by a ship 2.5 km away from the foghorn. The speed of sound in air is 330 m / s.

(i) Show that the wavelength of the sound is approximately 0.9 m.
State any equation you use in words or symbols.

[2]

(ii) Calculate the time it takes for sound to travel to the ship from the foghorn.

time = ... [2]

[Total: 10]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

5 A ray of light is incident on a soap film. Fig. 5.1 shows a magnified image of a small part of the soap film. The ray of light is refracted as it enters the soap film.

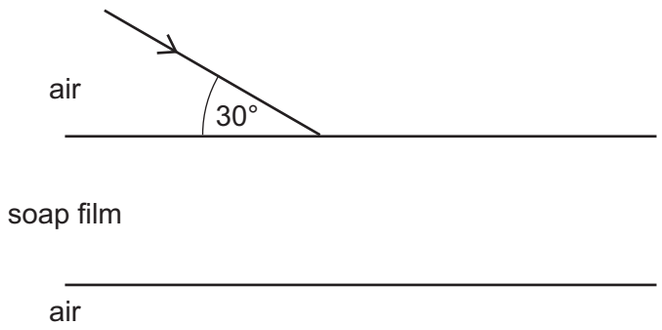


Fig. 5.1

The refractive index of the soap film is 1.28.

(a) Define refractive index in terms of the speed of light.

.....

..... [1]

(b) **(i)** Show that the angle of refraction as the light enters the soap film is approximately 43°.

[2]

(ii) On Fig. 5.1, carefully draw the refracted light ray in the soap film and label the angle of refraction. [2]

(c) The ray of light is monochromatic red light with a wavelength of 680 nm in air.

(i) Define monochromatic.

.....

..... [1]

(ii) Calculate the frequency of the light.

frequency = ... [3]

[Total: 9]

5 (a) (i) Electromagnetic waves have many uses.

On Fig. 5.1 draw **one** line from each use to the region of the electromagnetic (e-m) spectrum it uses.

Use	Region of e-m spectrum
security marking	gamma rays
Bluetooth	ultraviolet
optical fibres	infrared
detection of cancer	radio waves

Fig. 5.1

[2]

(ii) State **one** advantage of using microwaves compared with radio waves to transmit mobile (cell) phone signals.

.....

..... [1]

(b) Describe the difference between transverse waves and longitudinal waves.

.....

.....

.....

..... [2]

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

© UCLES 2025

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(c) Fig. 5.2 shows a simplified diagram of seismic P-waves travelling through the Earth. The paths of the waves are curved.
Seismic waves are produced by rock movements in earthquakes.

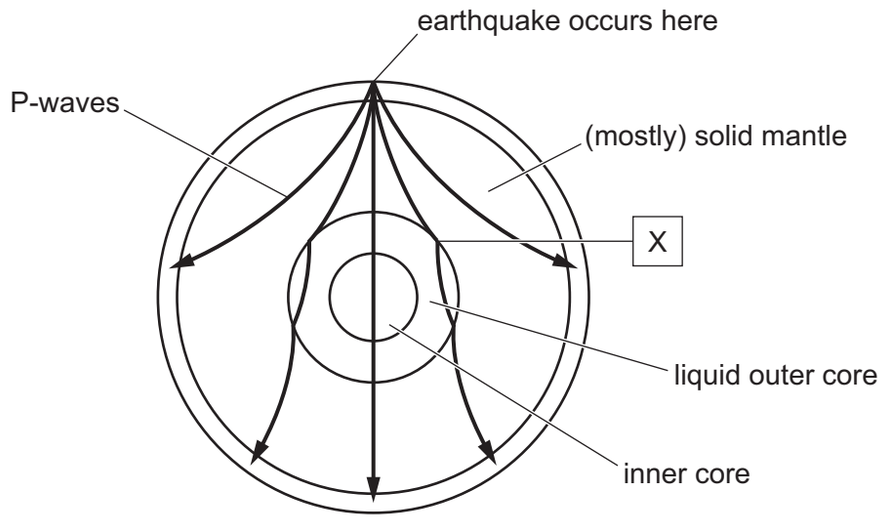


Fig. 5.2 (not to scale)

(i) State the type of wave that P-waves can be modelled as.

..... [1]

(ii) At point X, the P-wave travels from the solid mantle to the liquid core. There is a sudden change in direction of the path of the wave.

Explain the change in direction of the P-wave at point X.

.....

.....

..... [2]

(d) The Earth's surface absorbs incoming radiation from the Sun and also emits thermal radiation from its surface.

Over the past 50 years, scientists estimate that the average temperature of the Earth's surface has increased by approximately 0.75 °C.

Explain what is causing this average temperature rise.

.....

.....

.....

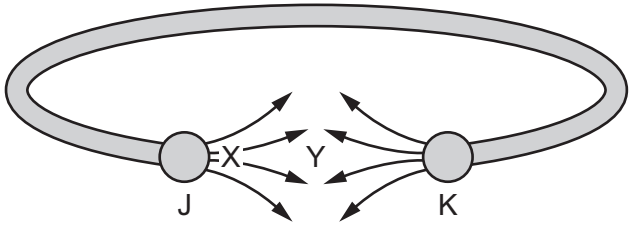
..... [2]

[Total: 10]

4 ELECTRICITY AND MAGNETISM

13 questions · Paper 2 & Paper 4

24 The diagram shows a bracelet containing two magnets. The arrows represent the pattern and the direction of the magnetic field due to the magnets.



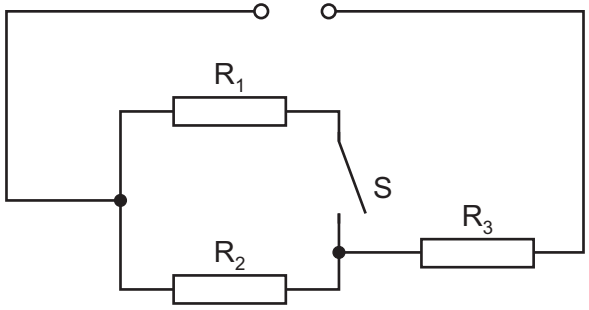
Which row identifies the magnetic pole at J, the magnetic pole at K and where the magnetic field is the strongest?

	magnetic pole at J	magnetic pole at K	where the magnetic field is the strongest
A	N	N	X
B	N	N	Y
C	S	S	X
D	S	S	Y

25 Which row describes an electric field and its direction?

	description of electric field	direction of electric field
A	region in which an electric charge experiences a force	direction of the force on a positive charge
B	region in which an electric charge experiences a force	direction of the force on a negative charge
C	region in which an electric charge produces a current	direction of the force on a positive charge
D	region in which an electric charge produces a current	direction of the force on a negative charge

27 The diagram shows three 20 Ω resistors, R_1, R_2 and R_3, and a switch S connected to a d.c. power supply.



Switch S is open.

The potential difference (p.d.) across R_3 is 9.0 V.

What is the p.d. across R_3 when switch S is closed?

A 3.0 V **B** 6.0 V **C** 12 V **D** 27 V

29 An electrical appliance has a symbol on it to indicate it is double-insulated.

Which statement is correct?

A A two-core, flexible cable with **no** earth wire is used to connect it to the mains supply.

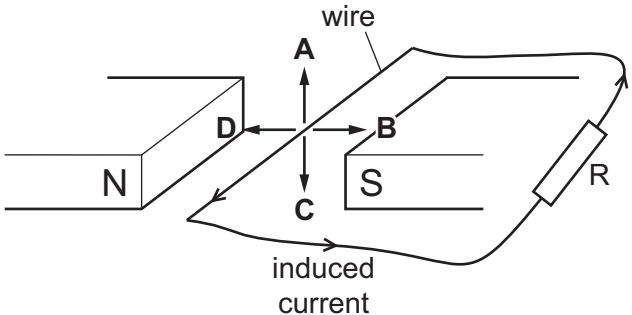
B It needs an earth wire that is connected to a fuse.

C **No** fuse or circuit breaker is needed.

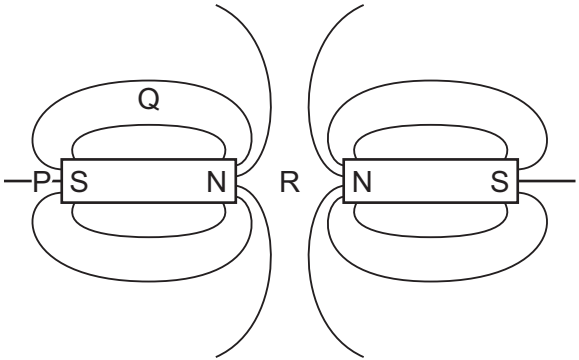
D The cable should be connected to the live and earth connectors of the plug but **not** the neutral.

29 A wire connected to a resistor R is moved in a magnetic field. A current is induced in the direction shown.

In which direction is the wire moved?



24 The diagram shows part of the magnetic field around two magnets.



Which list gives the order of magnetic field strengths at the positions P, Q and R, from weakest to strongest?

A $P \rightarrow Q \rightarrow R$

B $P \rightarrow R \rightarrow Q$

C $Q \rightarrow P \rightarrow R$

D $R \rightarrow Q \rightarrow P$

26 Copper wire is available in fixed lengths but in various diameters *d*. Each wire has a different resistance *R*.

Which relationship between *R* and *d* is correct?

A *R* is directly proportional to *d*.

B *R* is directly proportional to d^2.

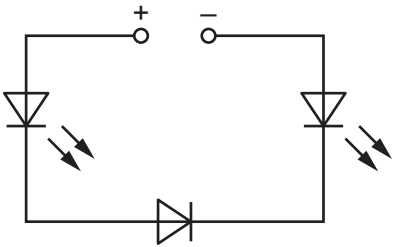
C *R* is inversely proportional to *d*.

D *R* is inversely proportional to d^2.

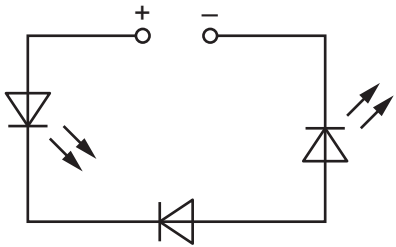
28 The circuits shown all include a d.c. power supply, one diode and two light-emitting diodes (LEDs).

In which circuit will both LEDs be turned on?

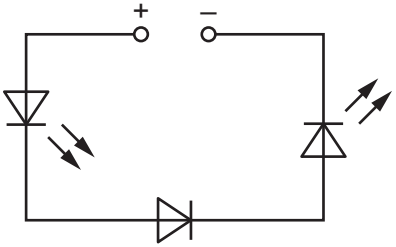
A



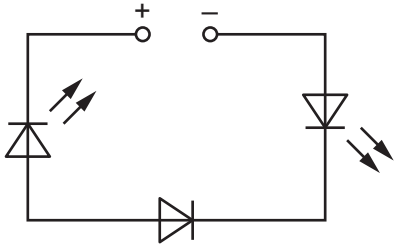
B



C



D



29 In which mains electric wires are the fuse and the switch placed?

	fuse	switch
A	earth	live
B	live	live
C	live	neutral
D	neutral	neutral

30 Which component is **not** found in an a.c. generator?

A coil of wire

B split-ring commutator

C magnetic poles

D slip rings

6 Fig. 6.1 shows a person using a magnetic window cleaner. The part on the outside of the window is attracted to the inside part through the glass window.

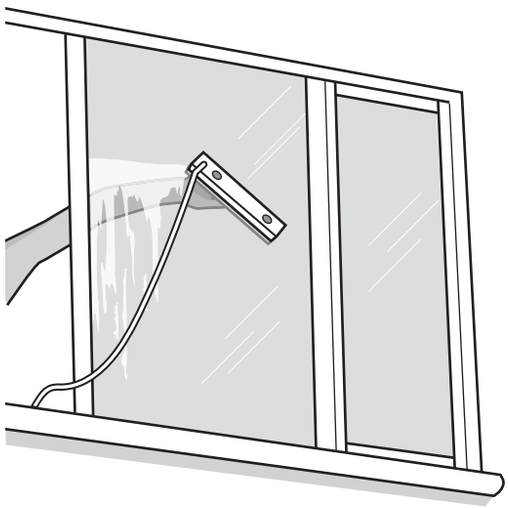


Fig. 6.1

Each part of the window cleaner contains two magnets. Fig. 6.2 shows the magnetic field between the parts of the window cleaner.

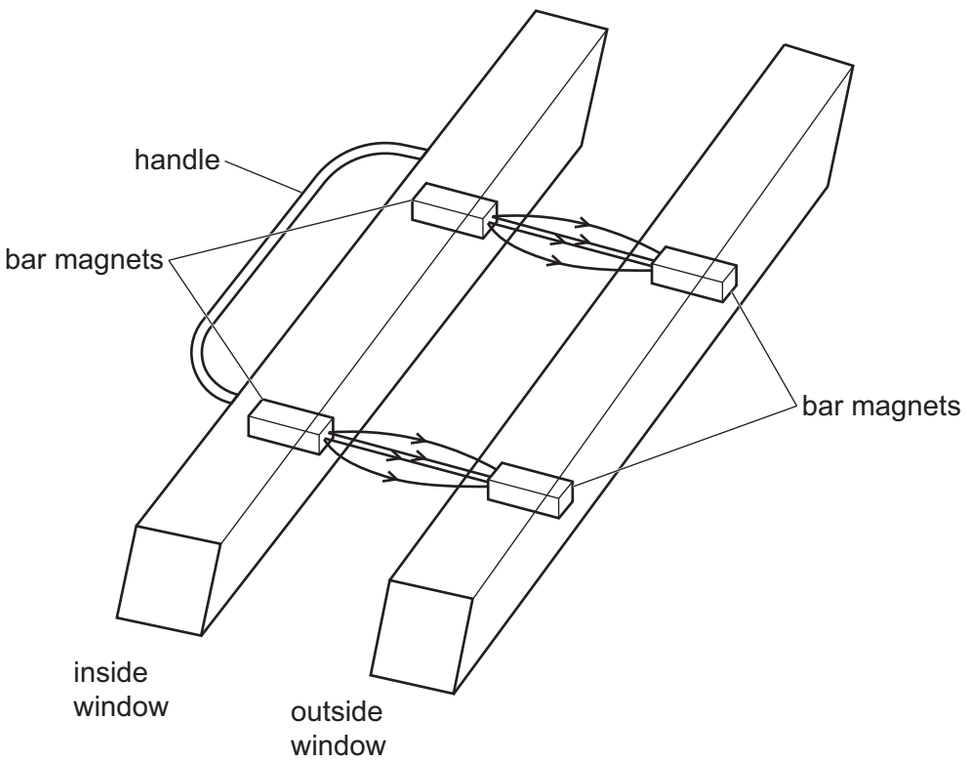


Fig. 6.2

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(a) Glass is **not** a magnetic material.

State the difference between magnetic and non-magnetic materials.

.....

..... [1]

(b) Suggest a suitable material for the magnets in the window cleaner. Explain your answer.

.....

..... [1]

(c) Label the poles of the magnets in Fig. 6.2. [1]

(d) State how the field lines in Fig. 6.2 show different strengths of the magnetic field between the magnets.

.....

..... [1]

[Total: 4]

* 0000800000006 *

3 Fig. 3.1 shows a mains electric heater used to heat a small room.

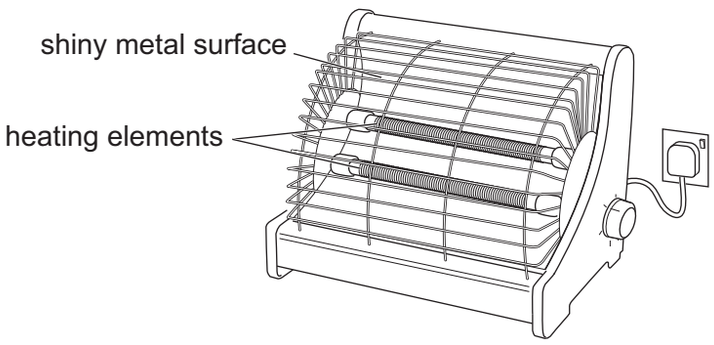


Fig. 3.1

(a) State the region of the electromagnetic spectrum which radiates thermal energy from the heater.

..... [1]

(b) Explain why the shiny metal surface behind the heating elements increases the thermal energy radiated into the room.

.....

.....

..... [2]

(c) The metal outer casing of the heater is earthed. State why this is an important safety feature.

.....

..... [1]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

© UCLES 2025

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(d) The mains voltage is 230 V. The two identical heating elements are connected in parallel. Each heating element has a resistance of 89 Ω.

(i) Calculate the current in **one** heating element.

current = ... [2]

(ii) Show that the electrical power of the heater is approximately 1200 W. State any equation you use in words or symbols.

[2]

(iii) The heater is 95% efficient at converting electrical work done to thermal energy.

Calculate the thermal energy emitted by the heater in **(d)(ii)** in 60 s. Give your answer to **two** significant figures.

thermal energy = ... [3]

[Total: 11]

7 **(a)** Fig. 7.1 shows a sketch of the current–voltage graph for an electrical component.

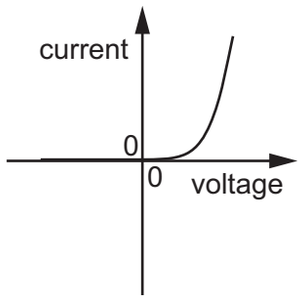


Fig. 7.1

(i) Name the electrical component.

Explain how you identified the component from the graph in Fig. 7.1.

name .....

explanation .....

.....

[2]

(ii) Draw the circuit symbol for this component.

[1]

(b) Fig. 7.2 shows an electric circuit for two identical electric heaters, A and B, connected to a mains supply of 230 V.

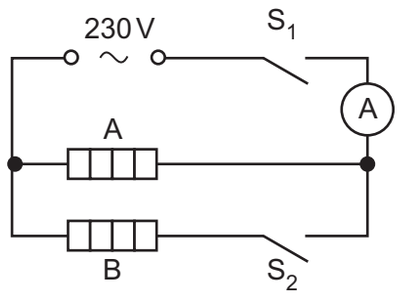


Fig. 7.2

S_1 is closed. S_2 is open. The reading on the ammeter is 3.9 A.

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

© UCLES 2025

DO NOT WRITE IN THIS MARGIN

(i) Calculate the resistance of heater A.

resistance = ... [2]

(ii) Calculate the energy transferred by heater A in 5.0 minutes.

energy = ... [3]

(iii) S_1 remains closed and S_2 is closed.

Determine the reading on the ammeter. Show your working.

ammeter reading = ... [2]

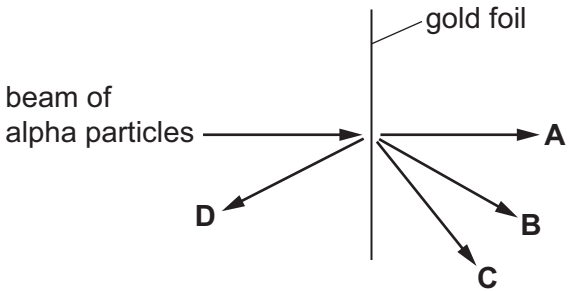
[Total: 10]

5 NUCLEAR PHYSICS

13 questions · Paper 2 & Paper 4

33 In an alpha particle scattering experiment, a beam of alpha particles is fired at a very thin piece of gold foil.

In which direction do most of the alpha particles travel after hitting the foil?



35 A detector placed near a radioactive isotope gives a measured count rate of 800 counts / minute. The half-life of the isotope is three hours.

The background count rate is 32 counts / minute.

What is the count rate on the detector after six hours?

A 192 counts / minute

B 200 counts / minute

C 224 counts / minute

D 232 counts / minute

34 An atom of boron contains 5 protons and 5 electrons. The nucleon number of the atom is 11.

How many neutrons are there in the atom?

A 5 **B** 6 **C** 11 **D** 16

36 Sodium-24 is a radioactive isotope that emits beta radiation.

Which nuclear equation shows how sodium-24 decays?

A $^{24}_{11}\text{Na} \rightarrow {}^{24}_{12}\text{Mg} + {}^{0}_{-1}\beta$

B $^{24}_{11}\text{Na} \rightarrow {}^{24}_{10}\text{Ne} + {}^{0}_{-1}\beta$

C $^{24}_{11}\text{Na} \rightarrow {}^{20}_{9}\text{F} + {}^{4}_{2}\beta$

D $^{24}_{11}\text{Na} \rightarrow {}^{20}_{13}\text{A}l + {}^{4}_{2}\beta$

33 $^{239}_{94}Pu$ is an isotope of plutonium.

The nucleus can absorb a neutron and then split to form two other nuclei.

Two nuclei that form are gold, $^{207}_{79}Au$, and phosphorus, $^{31}_{15}P$.

In this reaction, neutrons are also produced.

What is the number of neutrons produced when the nucleus splits?

A 1 **B** 2 **C** 4 **D** 5

37 Which safety precautions must be taken when using a source of gamma radiation?

1 Reduce the distance between the source and the person.

2 Reduce the time of exposure to the radiation.

3 Use a suitable shielding material between the source and the person.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

34 When a thin gold foil is bombarded with alpha particles, some alpha particles are deflected through large angles.

Which statement explains this deflection?

A Most of the atom consists of empty space.

B All of the positive charge and most of the mass of the gold atom are concentrated in a small volume.

C Positive charge in the gold atom is spread evenly throughout the atom.

D All of the negative charge is concentrated at its centre.

34 Three statements about alpha (α) particles and beta (β) particles are listed.

1 The magnitude of the charge on an α-particle is twice the charge on a β-particle.

2 The mass of an α-particle is large compared with the mass of a β-particle.

3 The kinetic energy of an α-particle is small compared to the kinetic energy of a β-particle.

Which statements help to explain the different ionising effects of α-particles and β-particles?

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

33 The arrow shows the path of an α-particle travelling towards the centre of a gold nucleus.



Which description of the path of the α-particle after striking the gold nucleus is correct?

A The α-particle is deflected through a very small angle towards the top of the page.

B The α-particle is deflected back along its initial path.

C The α-particle carries on in a straight line with **no** deflection.

D The α-particle is captured by the nucleus and goes into orbit around it.

35 A beta-particle is a fast-moving electron.

Which statement explains how beta-particles are emitted from an atom?

A An electron is emitted as a beta-particle from an inner electron shell of the atom.

B An electron is emitted as a beta-particle from an outer electron shell of the atom.

C A neutron changes into a proton and a beta-particle is emitted from the nucleus.

D A proton changes into a neutron and a beta-particle is emitted from the nucleus.

9 **(a)** Describe the structure of an atom of helium-4, $^{4}_{2}He$.

.....

.....

.....

..... [3]

(b) The Sun is a medium-sized star powered by nuclear fusion reactions which release energy.

(i) State what happens during nuclear fusion reactions which form helium.

.....

.....

..... [2]

(ii) State **two** regions of the electromagnetic spectrum by which the Sun radiates most of its energy.

..... [2]

(c) Describe what happens to a star when most of the fuel in its centre has been converted to helium.

.....

.....

..... [2]

[Total: 9]

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

9 Strontium-90 ($^{90}_{38}Sr$) is a radioactive isotope that contains 38 protons and 52 neutrons. Strontium-90 decays to form an isotope of yttrium (Y) by emitting beta (β) particles.

(a) (i) Suggest how the nucleus of a stable isotope of strontium differs from a nucleus of strontium-90. Explain your answer.

suggestion .....

explanation .....

.....

[2]

(ii) Complete the nuclide equation for the decay of strontium-90 to yttrium.

$$^{90}_{38}Sr \longrightarrow {}^{.....}_{.....}Y + {}^{.....}_{.....}\beta$$

[2]

(iii) Explain why scientists limit the amount of time they are exposed to radioactive strontium.

.....

..... [2]

(b) Yttrium is also unstable. A scientist places a sample of yttrium near a radiation detector. Table 9.1 shows the count rate recorded by the detector as the sample decays.

Table 9.1

time / h	recorded count rate / counts / min
0	68
50	49
100	38
150	32
200	26
250	24
300	20
350	21
400	20

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

Fig. 9.1 shows a graph of the count rate **due to yttrium** against time.

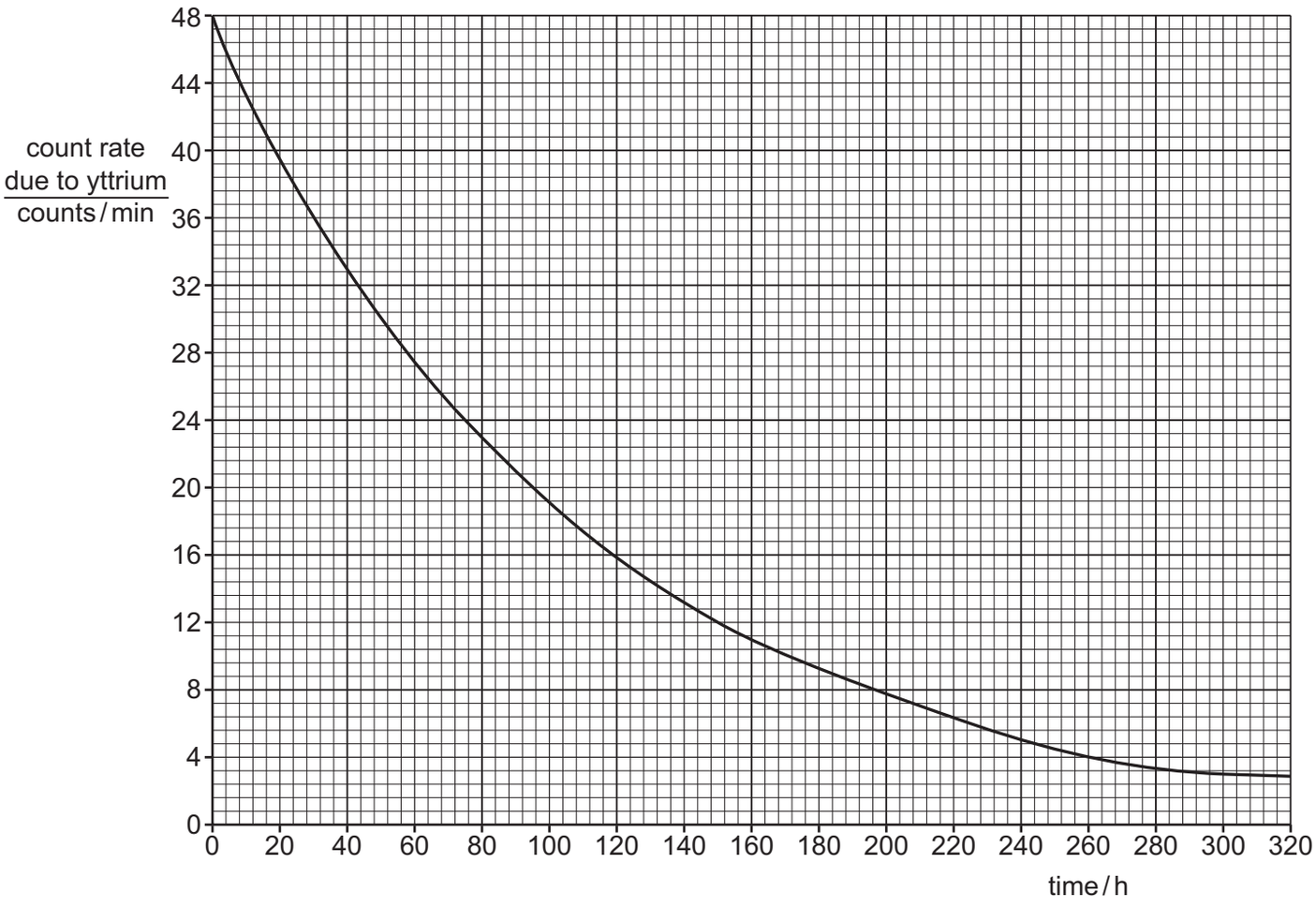


Fig. 9.1

(i) Use Fig. 9.1 to determine the half-life of yttrium. Show your working.

half-life = ... h [3]

(ii) Explain the difference between the count rate in Table 9.1 and the count rate due to yttrium plotted on the graph in Fig. 9.1.

.....

.....

..... [1]

[Total: 10]

© UCLES 2025

* 0000800000016 *

9 Strontium-90 is a radioactive isotope of strontium. The nuclide notation for strontium-90 is:

$^{90}_{38}Sr$

(a) (i) Explain what isotopes are.

.....

..... [1]

(ii) Complete Table 9.1 for strontium-90.

Table 9.1

particle	number in one atom	location
	38	outside nucleus
neutron		
	38	inside nucleus

[2]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

© UCLES 2025

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(b) Strontium-90 is used to measure the thickness of metal sheets in industry.
Strontium-90 decays by emitting beta (β) particles which pass through a metal sheet to a detector.

(i) One metal sheet is 0.75 mm thick. Suggest why strontium-90 is a suitable radioactive source to measure the thickness of the metal sheets.

.....

.....

..... [2]

(ii) The half-life of strontium-90 is approximately 27 years. Fig. 9.1 shows the shape of a decay curve.

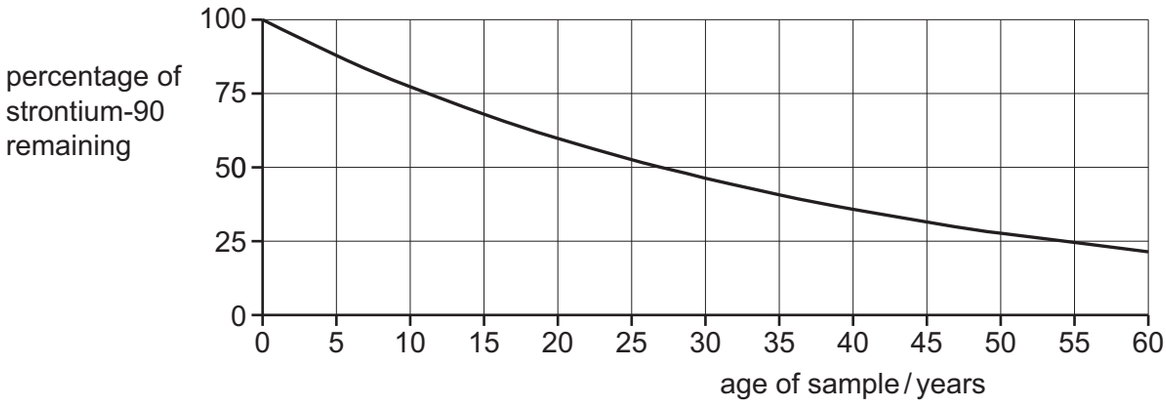


Fig. 9.1

The strontium-90 source is replaced with a new source after 15 years. Using Fig. 9.1, suggest why a strontium-90 source that is more than 15 years old needs to be replaced with a new source.

.....

..... [2]

[Total: 7]

6 SPACE PHYSICS

13 questions · Paper 2 & Paper 4

38 A student defines three periods of time:

- the time T_S taken for the Earth to orbit the Sun
- the time T_R taken for the Earth to rotate once on its axis
- the time T_M taken for the Moon to orbit the Earth.

Which row shows the shortest period of time and the longest period of time?

	shortest period of time	longest period of time
A	T_S	T_R
B	T_R	T_S
C	T_R	T_M
D	T_M	T_S

39 Which statement about the Universe is correct?

A The Universe consists of 100 000 galaxies.

B Redshift is a decrease in the observed wavelength of electromagnetic radiation emitted from receding galaxies.

C Light emitted from distant galaxies appears redshifted.

D Redshift of the light from distant galaxies gives evidence that the Universe is contracting.

38 A satellite is in orbit around the Earth. The orbit has radius r and period T.

Which expression gives the orbital speed of the satellite?

A $\frac{r}{2\pi T}$ **B** $\frac{T}{2\pi r}$ **C** $\frac{2\pi r}{T}$ **D** $\frac{2\pi T}{r}$

40 Which statement about the cosmic microwave background radiation (CMBR) is correct?

A CMBR now has a frequency higher than that of infrared radiation.

B CMBR has always had exactly the same frequency.

C CMBR had a shorter wavelength in the past.

D CMBR was produced a long time after the Universe was formed.

39 Comets move round the Sun.

Which statement describes the orbit of a comet?

A The orbit is circular with the Sun at the centre of the circle.

B The orbit is circular with the Sun **not** at the centre of the circle.

C The orbit is elliptical with the Sun at the centre of the ellipse.

D The orbit is elliptical with the Sun **not** at the centre of the ellipse.

40 A galaxy is at a distance of 60 million light-years from the Earth. The galaxy is moving away at a speed of 1300 km / s.

One light-year is about 9.5×10^{12} km.

There are about 3.2×10^{7} seconds in one year.

Which expression can be used to estimate the age of the Universe in years?

A $\dfrac{60 \times 10^{6} \times 3.2 \times 10^{7}}{1300 \times 9.5 \times 10^{12}}$

B $\dfrac{60 \times 10^{6} \times 9.5 \times 10^{12}}{1300 \times 3.2 \times 10^{7}}$

C $\dfrac{1300 \times 9.5 \times 10^{12}}{60 \times 10^{6} \times 3.2 \times 10^{7}}$

D $\dfrac{1300 \times 3.2 \times 10^{7}}{60 \times 10^{6} \times 9.5 \times 10^{12}}$

38 Which statement is correct?

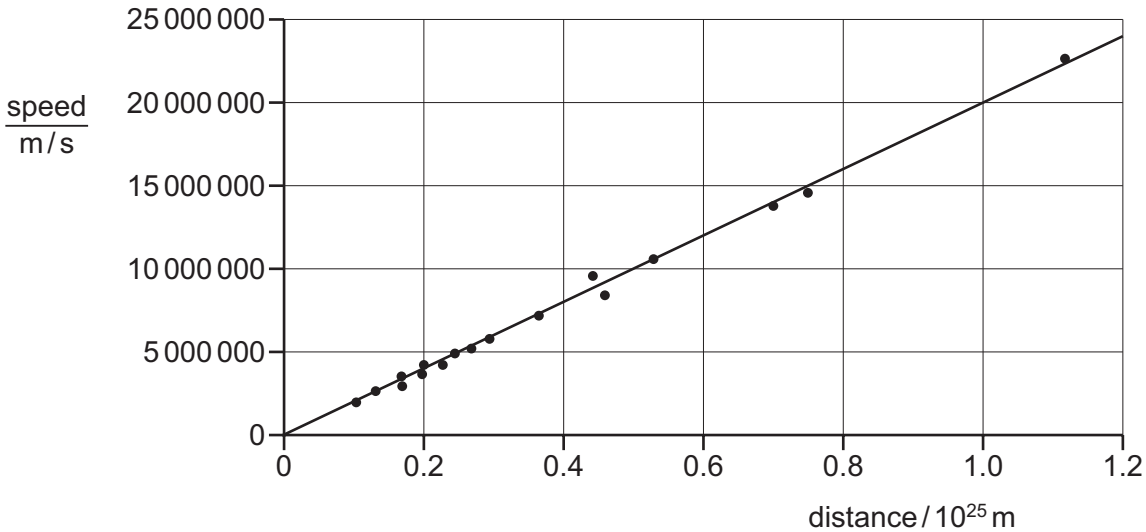
A The Earth orbits the Sun once in approximately 24 hours.

B The Earth orbits the Moon once in approximately 1 week.

C The Moon orbits the Earth once in approximately 1 month.

D The Sun orbits the Earth once in approximately 365 days.

40 A student plots a graph of the speed of different galaxies moving away from the Earth against their distance from the Earth.



Using the best straight line shown on the graph, what is the value of the Hubble constant H_0?

A $2.0 \times 10^{-18}\,s^{-1}$

B $2.2 \times 10^{-18}\,s^{-1}$

C $4.5 \times 10^{17}\,s$

D $5.0 \times 10^{17}\,s$

38 In which direction does the Sun appear to move every day, when viewed from the Earth?

A east to west

B north to south

C south to north

D west to east

39 Stars are powered by nuclear reactions that release energy.

Which statement describes the nuclear reaction in stable stars?

A Nuclear fission converts helium into hydrogen.

B Nuclear fission converts hydrogen into helium.

C Nuclear fusion converts helium into hydrogen.

D Nuclear fusion converts hydrogen into helium.

10 Jupiter and the Earth are planets in our Solar System.

(a) Describe the composition of Jupiter and the Earth.

Jupiter .....

the Earth .....
[2]

(b) The gravitational field strength at the surface of the Earth is approximately 9.8 N / kg. The gravitational field strength at the surface of Jupiter is approximately 23 N / kg.

(i) Define gravitational field strength.

.....

..... [2]

(ii) State **one** factor which causes the difference between the gravitational field strength at the surface of Jupiter and the gravitational field strength at the surface of the Earth.

.....

..... [1]

(c) State and explain the difference between the orbital speed of Jupiter and the orbital speed of the Earth.

statement .....

explanation .....

.....
[3]

[Total: 8]

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of Cambridge Assessment. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which is a department of the University of Cambridge.

© UCLES 2025

DO NOT WRITE IN THIS MARGIN DO NOT WRITE IN THIS MARGIN DO NOT WRITE IN THIS MARGIN DO NOT WRITE IN THIS MARGIN DO NOT WRITE IN THIS MARGIN

10 **(a)** Define a light-year in words.

..... [1]

(b) It takes light 490 s to travel from the Sun to the Earth.

Calculate the distance from the Sun to the Earth.

distance = ... [2]

(c) Fig. 10.1 is a scatter graph showing how the speed of galaxies moving away from the Earth varies with their distances from the Earth.

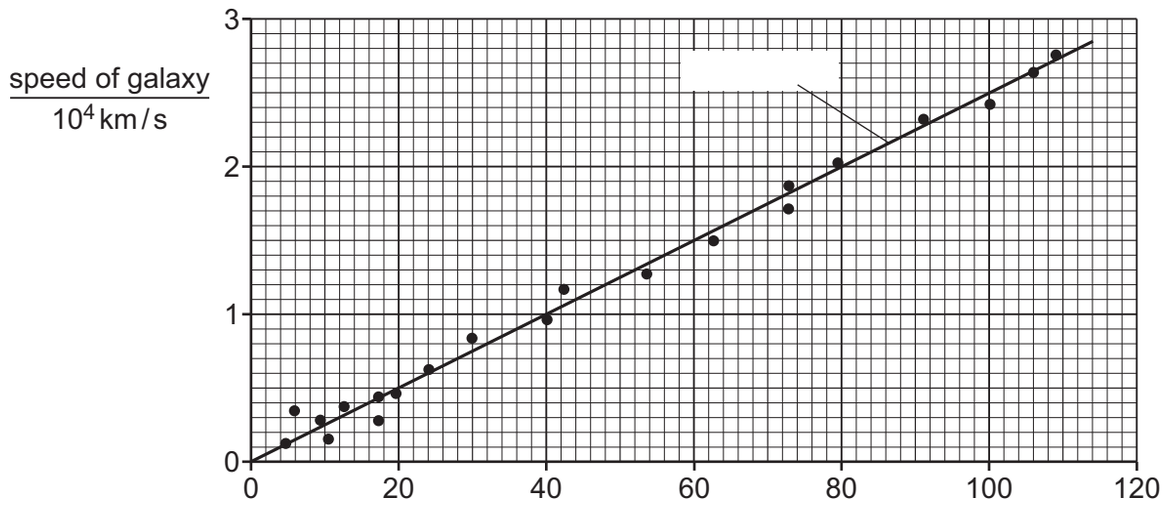


Fig. 10.1

A scientist draws a best-fit line on the scatter graph.

Use this best-fit line to determine a value for the Hubble constant.

Hubble constant = ... [3]

(d) The speed of a receding galaxy can be estimated using redshift.

Describe what is meant by redshift.

.....

.....

.....

..... [2]

© UCLES 2025

[Total: 8]

DO NOT WRITE IN THIS MARGIN DO NOT WRITE IN THIS MARGIN DO NOT WRITE IN THIS MARGIN DO NOT WRITE IN THIS MARGIN DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

10 Fig. 10.1 shows the orbit of the Earth and the orbit of a comet around the Sun.

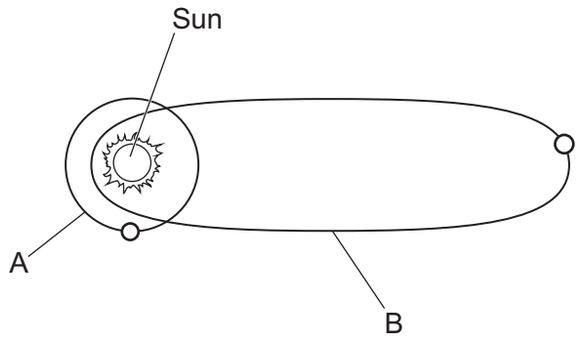


Fig. 10.1

(a) State which orbit, A or B, is the orbit of the comet.
Explain your answer.

orbit of comet is .....

explanation .....

.....
[1]

(b) Describe and explain how the motion of the comet changes as it orbits the Sun.

.....

.....

..... [2]

(c) At one position in its orbit, the comet is 6.6×10^{-6} light-years away from the Earth.

(i) State the meaning of light-year.

.....

..... [1]

(ii) Determine the distance in metres between the comet and the Earth.

distance = ... m [2]

[Total: 6]

© UCLES 2025